

Exercise 1

At admission to a college an entrance test is administered to 20 randomly selected students. The study aims to determine whether a student’s grade point average (GPA) at the end of the first year (y) can be predicted from the entrance test score (x). Assume that a Gaussian linear model $Y_i = \beta_1 + \beta_2 x_i + \varepsilon_i$ is fitted. The observed values of the entrance test and of the GPA for the students are:

unit	1	2	3	4	5	6	7	8	9	10
x	5.50	4.80	4.70	3.90	4.50	6.20	6.00	5.20	4.70	4.30
y	3.10	2.30	3.00	1.90	2.50	3.70	3.40	2.60	2.80	1.60
unit	11	12	13	14	15	16	17	18	19	20
x	4.90	5.40	5.00	6.30	4.60	4.30	5.00	5.90	4.10	4.70
y	2.00	2.90	2.30	3.20	1.80	1.40	2.00	3.80	2.20	1.50

From the data it is possible to compute the following useful quantities:

$\sum_{i=1}^{20} x_i = 100.00$ $\sum_{i=1}^{20} y_i = 50$ $\sum_{i=1}^{20} x_i^2 = 509.12$ $\sum_{i=1}^{20} y_i^2 = 134.84$ $\sum_{i=1}^{20} x_i y_i = 257.66$

Moreover, from fitting the model we know that

$$\widehat{var}(\hat{\beta}_1(Y)) = 0.7268^2 \qquad \widehat{var}(\hat{\beta}_2(Y)) = 0.1440^2$$
$$\sum_{i=1}^{20} (y_i - \hat{y}_i)^2 = 3.4063$$

- a) State the assumptions on ε_i , $i = 1, \dots, 20$.
- b) Obtain the maximum likelihood estimates of (β_1, β_2) and interpret them.
- c) Write the expression of the estimated regression function and plot it (in the figure below).
- d) Obtain a 0.99 confidence interval for β_2 . Does it include zero? Why might one be interested in whether the confidence interval includes zero?
- e) Is β_2 statistically significant using a significance level of 1%?
- f) Two new students “A” and “B” take the test. The result of the entrance test of student “A” is 5.0, the result of the entrance test of student “B” is 6.5. Imagine you want to estimate their mean GPA using a 0.95 confidence interval. What student do you expect to have the wider confidence interval? Why?

2

n=20 GAUSSIAN LINEAR MODEL

$$Y_i = \beta_1 + \beta_2 x_i + \varepsilon_i$$

- $E(\varepsilon_i) = 0 \quad \forall i = 1, \dots, n$
- $VAR(\varepsilon_i) = \sigma^2 > 0 \quad \forall i = 1, \dots, n$
- $Cov(\varepsilon_i; \varepsilon_k) = 0 \quad \text{for } i \neq k \rightarrow Cov(\varepsilon_i; \varepsilon_k) = 0$
- $\varepsilon_i \sim N(0, \sigma^2)$
- λ : FIXED AND NOT STOCHASTIC
- λ : NOT ALL EQUAL

6 MLE OF β_1 AND β_2

$$\hat{\beta}_2 = \frac{\sum x_i y_i}{\sum x_i^2} = \frac{\frac{1}{n-1} \sum (x_i - \bar{x})(y_i - \bar{y})}{\frac{1}{n-1} \sum (x_i - \bar{x})^2} = \frac{\sum x_i y_i - \bar{x} \bar{y} \cdot n}{\sum x_i^2 - n \bar{x}^2} = \frac{-n \bar{x} \bar{y} - n \bar{x} \bar{y} + n \bar{x} \bar{y} + \sum x_i y_i}{n \bar{x}^2 - 2n \bar{x}^2 + \sum x_i^2} = \frac{-n \bar{x} \bar{y} + \sum x_i y_i}{-n \bar{x}^2 + \sum x_i^2} = \frac{-20 \cdot \frac{100}{20} \cdot \frac{50}{20} + 257.66}{-20 \cdot (\frac{100}{20})^2 + 509.12} = \frac{7.66}{9.12} = 0.839$$

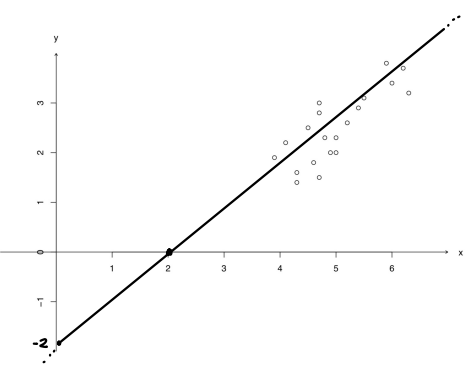
$$\hat{\beta}_1 = \bar{y} - \hat{\beta}_2 \bar{x} = \frac{50}{20} - 0.839 \cdot \frac{100}{20} = -1.699$$

• β_1 (COEFFICIENT) CAN NOT BE INTERPRETED IN A WAY THAT MAKE SENSE (ENTRANCE TEST SCORE $\rightarrow$) IS MEANINGLESS ^{ON AVERAGE} IS THE GPA CORRESPONDING TO A ENTRANCE SCORE OF 0
• ON AVERAGE, IF WE INCREASE THE ENTRANCE SCORE BY ONE UNIT, WE INCREASE GPA OF 0.839

7

$$Y_i = -1.699 + 0.839 x_i$$

$$Y = 0 \rightarrow -1.699 + 0.839 x \rightarrow x = 2.02$$



d

$1 - \alpha = 0.99 \rightarrow \alpha = 0.01 \rightarrow \alpha/2 = 5 \cdot 10^{-3} \rightarrow 1 - \alpha/2 = 0.995$

$$CI(\beta_2) = \left[\hat{\beta}_2 - t_{n-p; 1-\alpha/2} SE(\hat{\beta}_2); \hat{\beta}_2 + t_{n-p; 1-\alpha/2} SE(\hat{\beta}_2) \right] =$$
$$= \left[0.839 - 2.8734 \cdot 0.144; 0.839 + 2.8734 \cdot 0.144 \right] =$$
$$= \left[0.4145; 1.253 \right]$$

- NOT INCLUDE 0
- IF INTERVAL INCLUDE 0, THE ENTRANCE TEST SCORE MAY NOT BE STATISTICALLY SIGNIFICANT ON THE GPA (CHANGES IN TEST SCORE ARE NOT CORRELATED WITH GPA)
- ↓
- SINCE 0 NOT IN CONFIDENCE INTERVAL THEN THERE IS A SIGNIFICANT RELATIONSHIP BETWEEN TEST SCORE AND GPA

e USELESS WE REJECT $H_0 (\beta_2 = 0)$ SINCE 0 $\notin CI$

f

$x_A = 5.0$ WIDER RANGE STUDENT B SINCE IS MORE OUTLIER THAN A
 $x_B = 6.5$ BECAUSE IS FURTHER FROM THE MEAN $\bar{x} = \frac{100}{20} = 5$
 $1 - \alpha = 0.95$

$VAR(\hat{\beta}_2(Y)) = 0.1440^2 \rightarrow SE(\hat{\beta}_2) = 0.1440$

$t_{n-p; 1-\alpha/2} = t_{18; 0.995} = 2.8734$